



ASSIGNMENT 2 PRESENTATION

M11401023 林冠廷

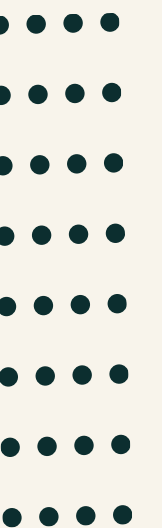
M11401102 陳喻宣

M11401204 溫翎傑



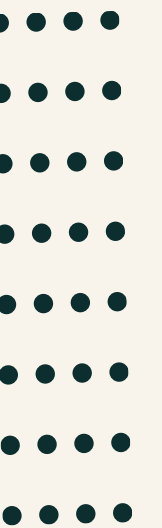
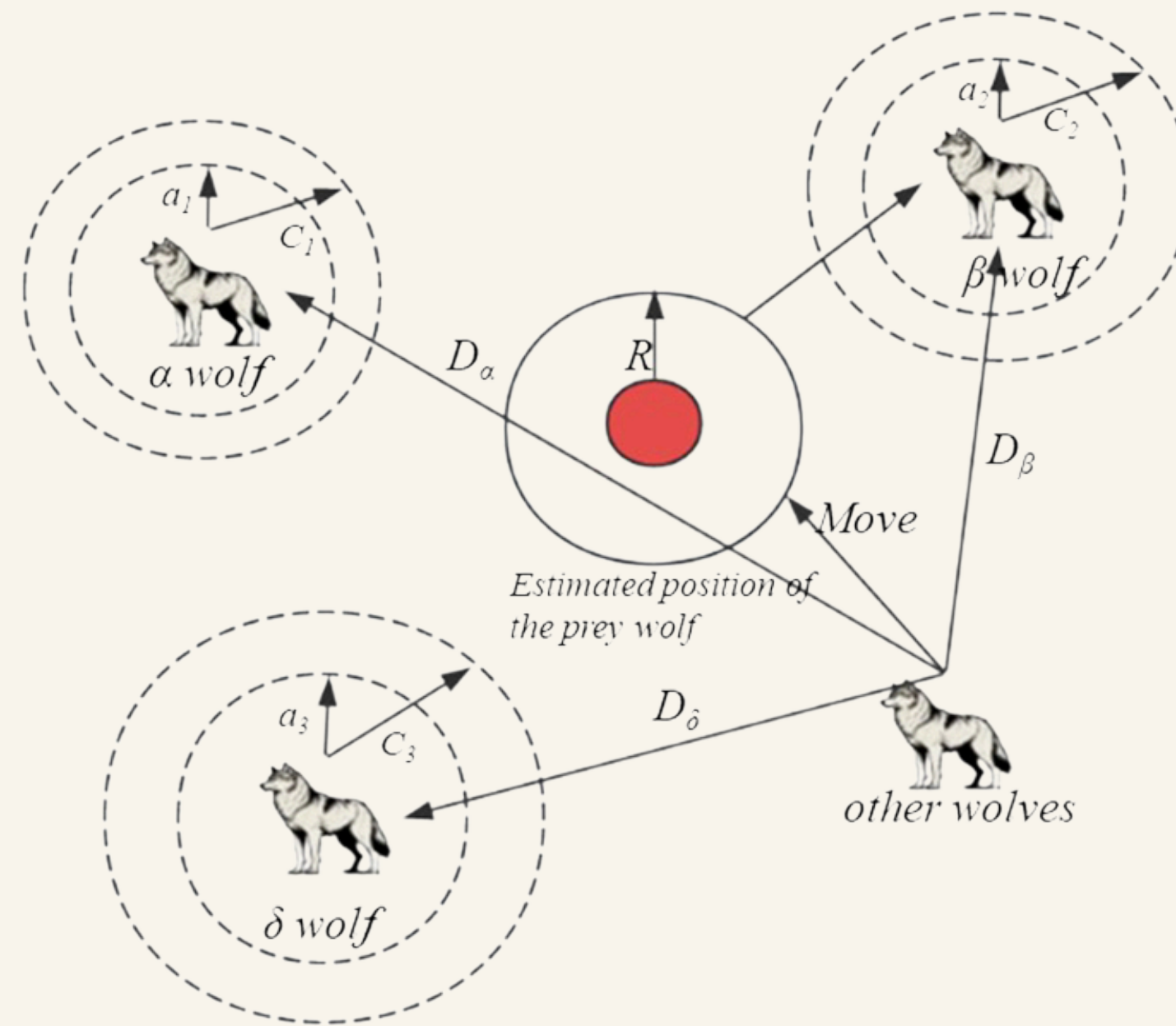
OUTLINE

1. GWO intro
2. GWO weakness and How to modify
3. Ideas for modified result

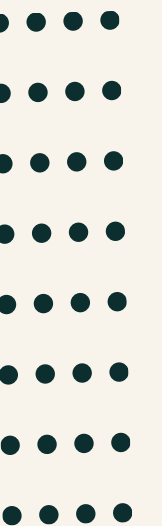
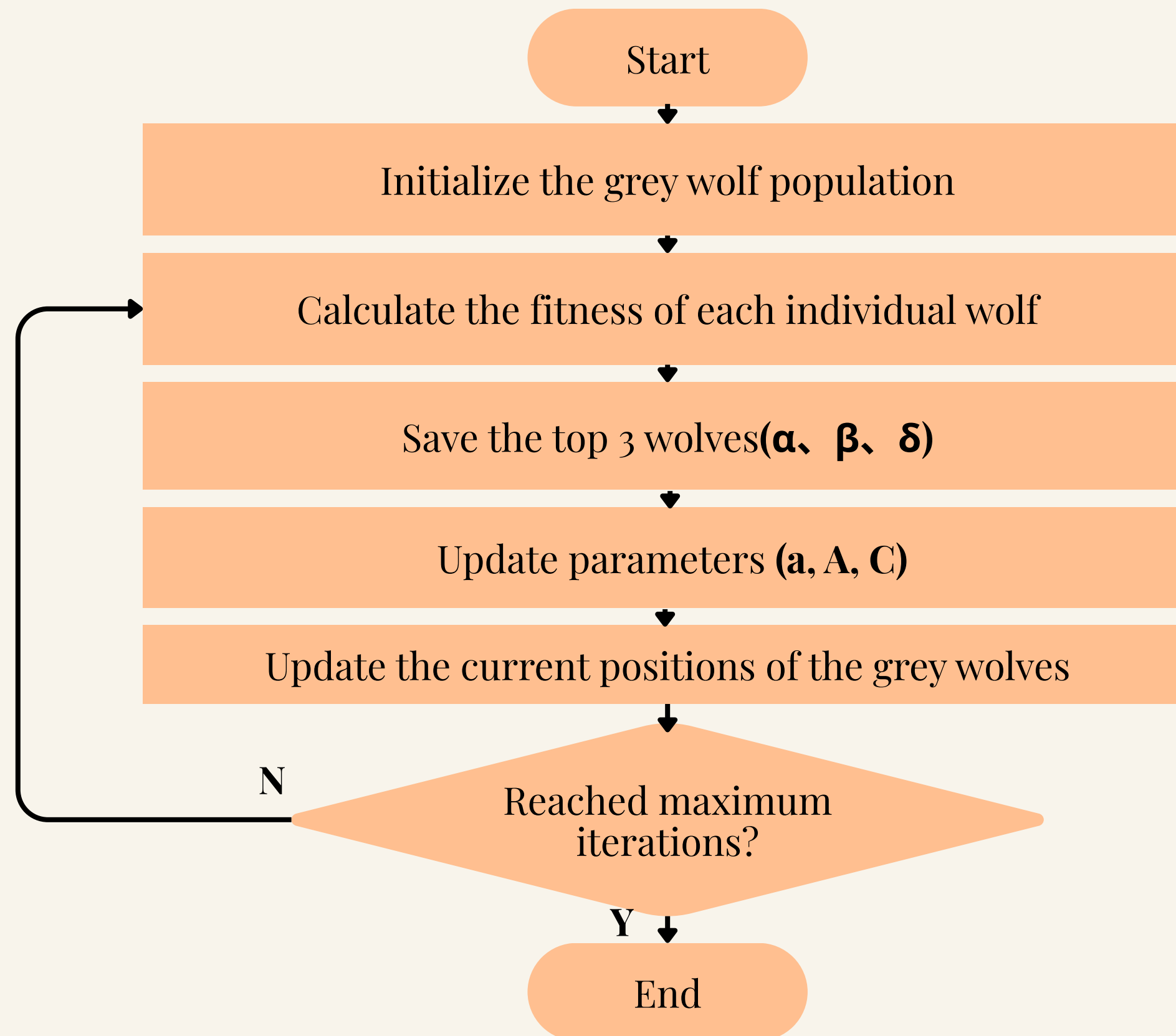


GREY WOLF OPTIMIZER, GWO

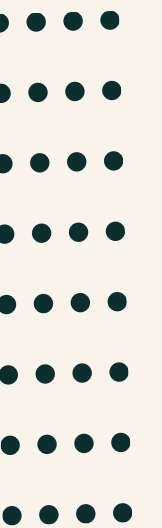
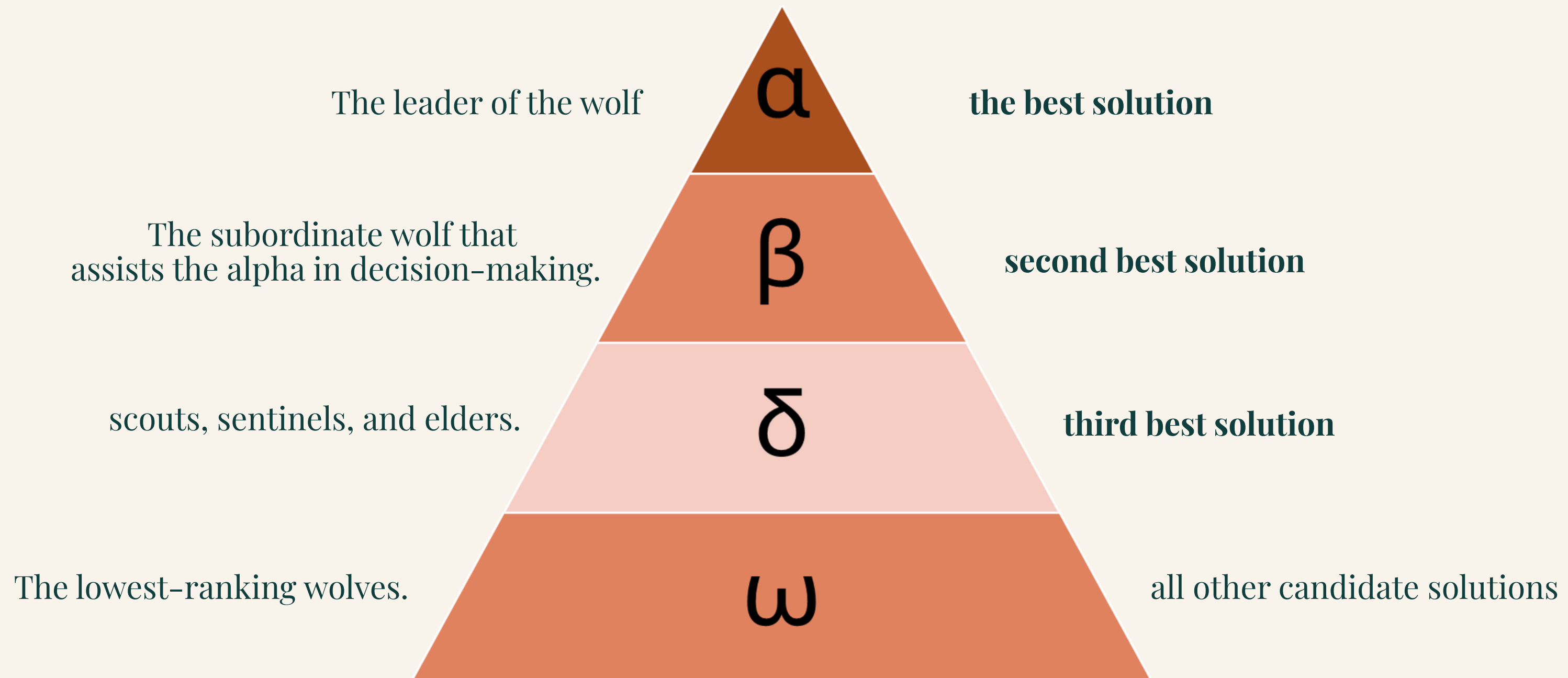
INTRODUCTION



GWO Algorithm Flowchart



SOCIAL HIERARCHY



Core Mechanisms

Encircling

$$D_i = |C_i \cdot X_i - X|, \quad i \in \{\alpha, \beta, \delta\}$$



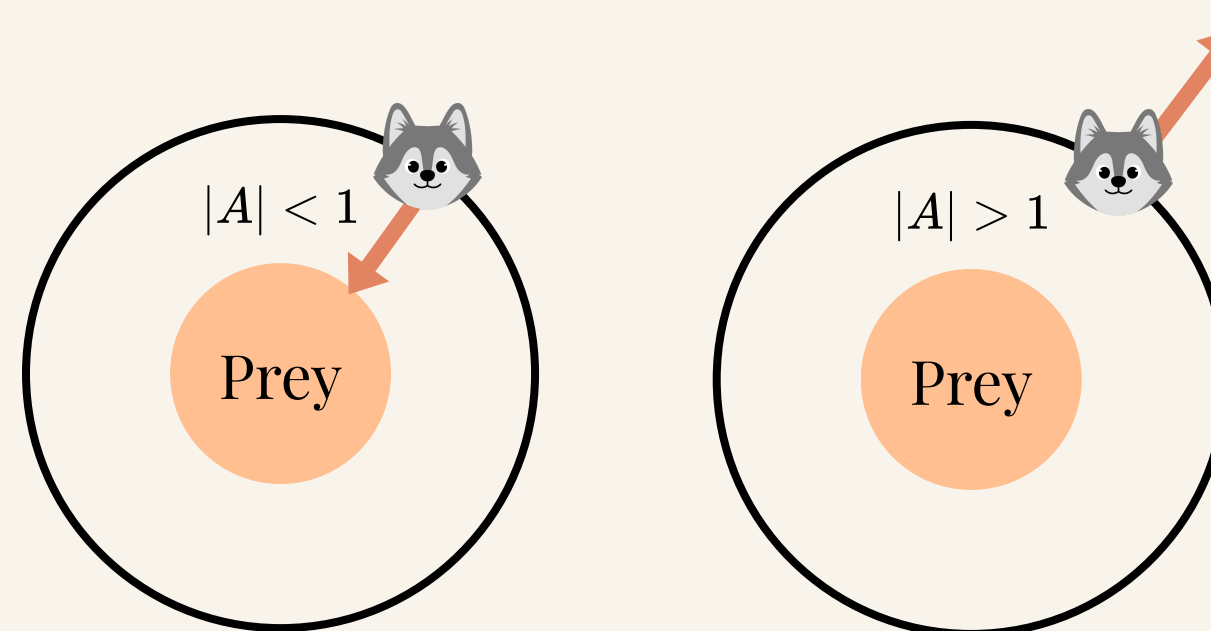
Hunting

$$X'_i = X_i - A_i \cdot D_i, \quad i \in \{\alpha, \beta, \delta\}$$

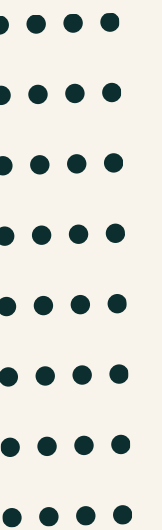
$$X(t+1) = \frac{X'_\alpha + X'_\beta + X'_\delta}{3}$$



Attacking

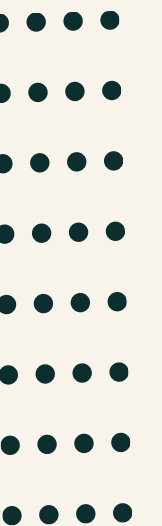


$$A \in [-a, a]$$





GWO WEAKNESS AND HOW TO MODIFY



How to improve the GWO?

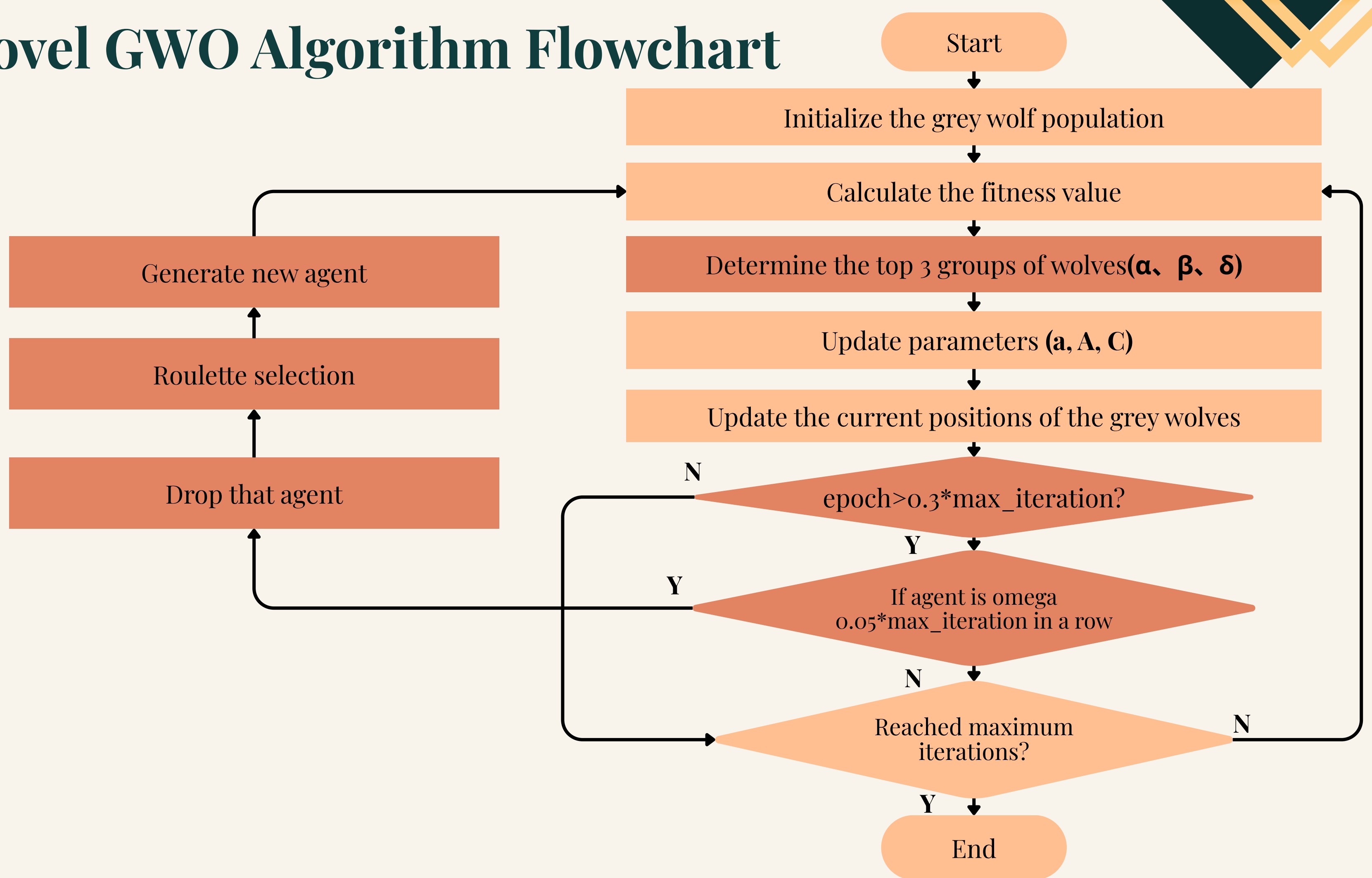
Social Hierarchy

Originally, alpha, beta, and delta each consisted of only one agent. This was modified into groups so that position updates do not rely solely on the top three agents (placing too much responsibility on them).

Wolves birth and death

ω can be considered ineffective, therefore, a birth–death mechanism is introduced to allow new agents the opportunity to escape local optima.

A Novel GWO Algorithm Flowchart



SOCIAL HIERARCHY

Population size = N

$$m = \lceil 0.1N \rceil$$

Top 3 best solution

α

The subordinate wolf that
assists the alpha in decision-making.

β

scouts, sentinels, and elders.

δ

The lowest-ranking wolves.

ω

$$G_{\alpha}(t) = \{\mathbf{X}_1(t), \dots, \mathbf{X}_m(t)\}$$

$$G_{\beta}(t) = \{\mathbf{X}_{m+1}(t), \dots, \mathbf{X}_{2m}(t)\}$$

$$G_{\delta}(t) = \{\mathbf{X}_{2m+1}(t), \dots, \mathbf{X}_{3m}(t)\}$$

Core Mechanisms

Encircling

$$\mathbf{D}_k^g = |\mathbf{C}_k \odot \mathbf{X}_k^g(t) - \mathbf{X}(t)|, \quad k \in \{\alpha, \beta, \delta\}$$

$$\mathbf{X}_\alpha^g = \arg \min_{\mathbf{X} \in G_\alpha} f(\mathbf{X}) \quad \mathbf{X}_\beta^g = \text{Unif}(G_\beta) \quad \mathbf{X}_\delta^g = \text{Unif}(G_\delta)$$



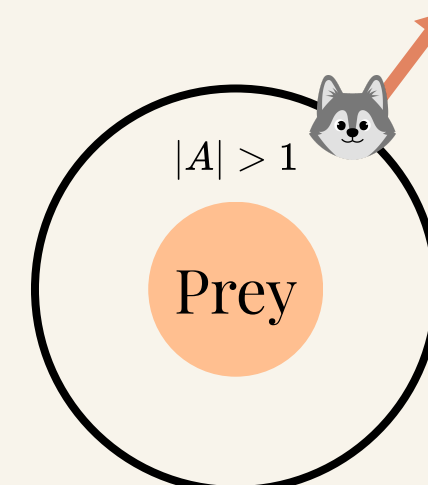
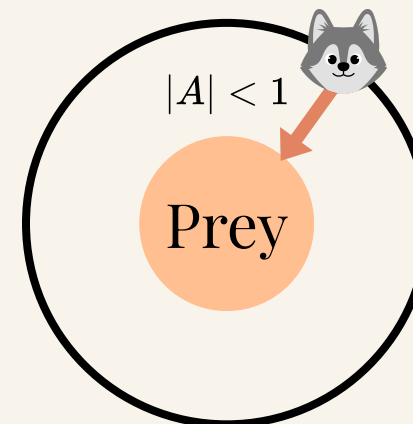
Hunting

$$\mathbf{X}'_k = \mathbf{X}_k^g(t) - \mathbf{A}_k \odot \mathbf{D}_k^g, \quad k \in \{\alpha, \beta, \delta\}$$

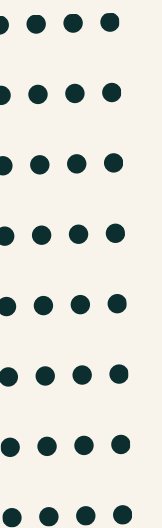
$$\mathbf{X}(t+1) = \frac{\mathbf{X}'_\alpha + \mathbf{X}'_\beta + \mathbf{X}'_\delta}{3}$$



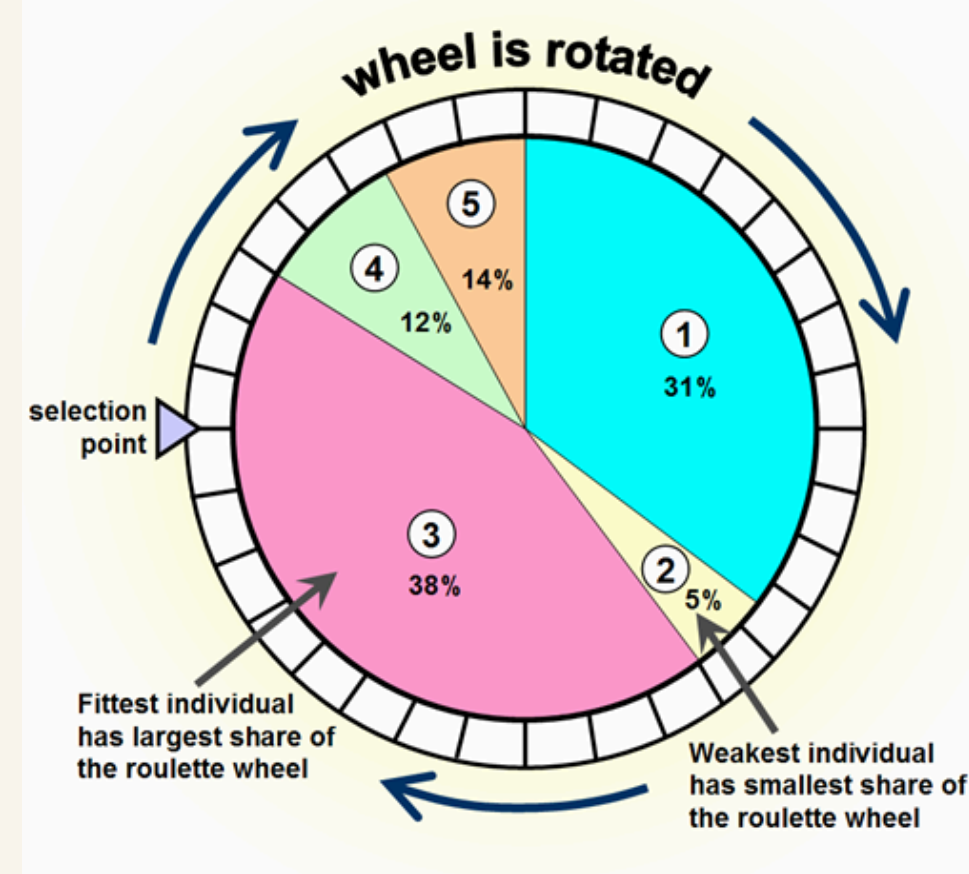
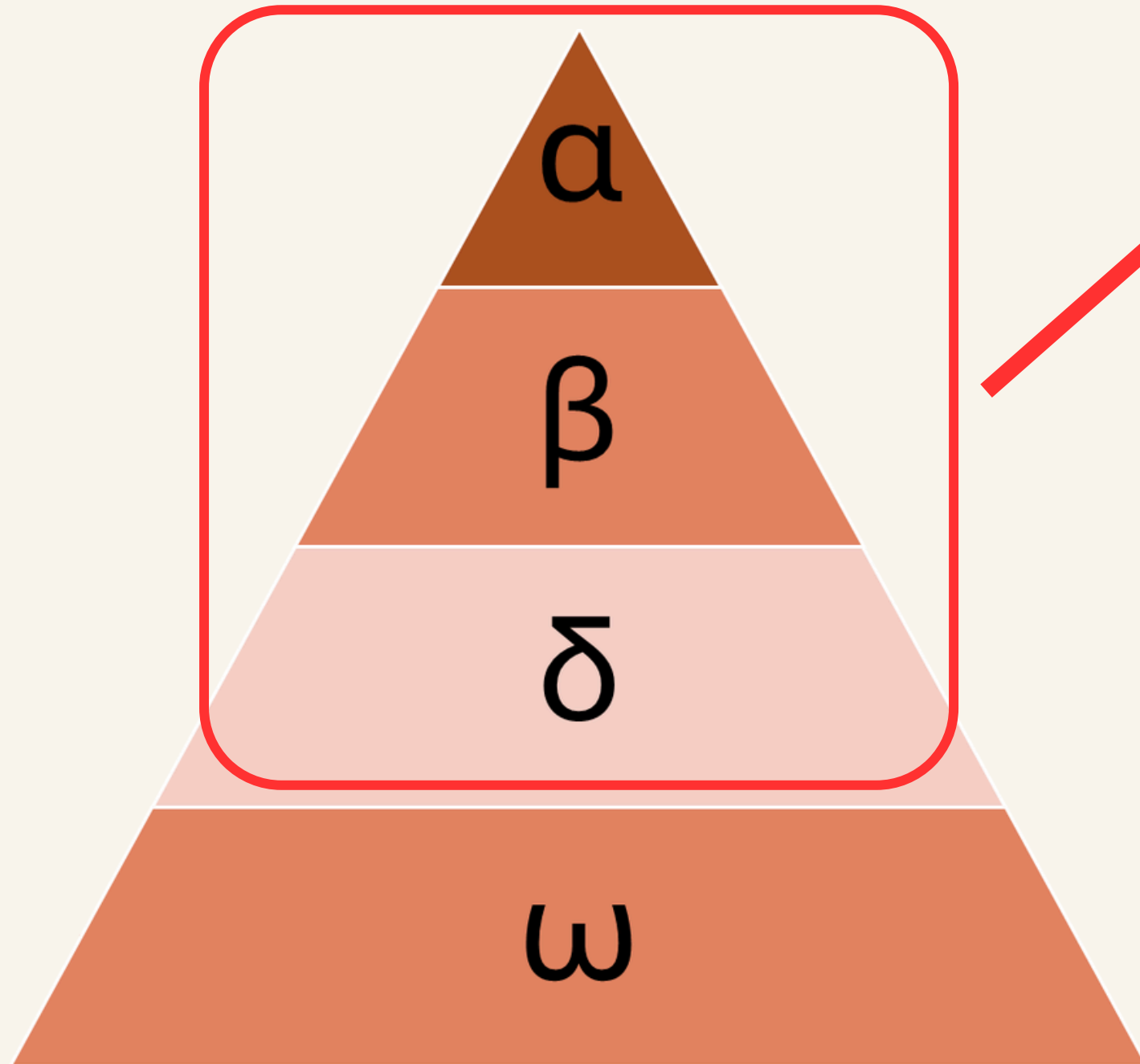
Attacking



$$A \in [-a, a]$$



BIRTH AND DEATH(LEVY LIGHT)



Roulette selection

Select 2 agents

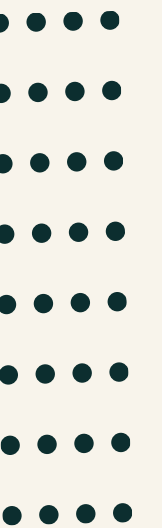
$$\mathbf{X}_{\text{base}}(t) = 0.5\mathbf{X}_{s_1}^g(t) + 0.5\mathbf{X}_{s_2}^g(t)$$

Levy Flight

$$\mathbf{X}_{\text{new}}(t) = \mathbf{X}_{\text{base}}(t) + \eta\mathbf{L}(t)$$



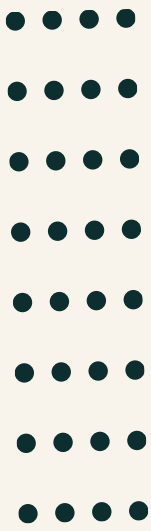
IDEAS FOR MODIFIED RESULT





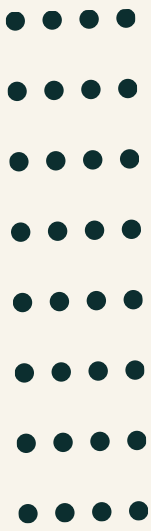
ABLATION EXPERIMENT-ORIGINAL

	Dim	Best	Mean	Worst	Succes rate(%)
Rastrigin	10	0	0.000001	0.000001	100
	30	0.000012	0.000017	0.000026	100
	50	0.000088	0.00011	0.000158	100
Styblinski-Tang	10	-391.661	-360.898	-329.741	10
	30	-936.37	-795.742	-678.122	0
	50	-1201.69	-1046.72	-948.905	0
-39.16617 * N					
Griewank	10	0	0.00494	0.03122	90
	30	0	0.00098	0.009796	100
	50	0	0.001345	0.013447	90



ABLATION EXPERIMENT-
GROUP(FIX ALPHA, RANDOM BETA DELTA)

	Dim	Best	Mean	Worst	Succes rate(%)
Rastrigin	10	0	0.000001	0.000002	100
	30	0.000054	0.000097	0.000158	100
	50	0.000317	0.00054	0.000762	100
Styblinski-Tang -39.16617 * N	10	-352.328	-324.111	-299.306	0
	30	-623.985	-587.258	-520.164	0
	50	-908.881	-847.778	-791.056	0
Griewank	10	0	0.006611	0.041062	90
	30	0	0	0	100
	50	0	0	0.000001	100





ABLATION EXPERIMENT-BORN DEATH

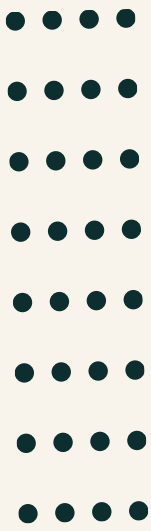
	Dim	Best	Mean	Worst	Succes rate(%)
Rastrigin	10	0	0.000001	0.000001	100
	30	0.000006	0.000019	0.000034	100
	50	0.000078	0.00012	0.000181	100
 Styblinski-Tang -39.16617 * N	10	-391.662	-370.128	-349.251	20
	30	-888.351	-818.265	-693.102	0
	50	-1219.16	-1092.22	-990.573	0
 Griewank	10	0	0.016154	0.043706	60
	30	0	0.00543	0.045318	90
	50	0	0.001268	0.012684	90





RESULT-HYBRID

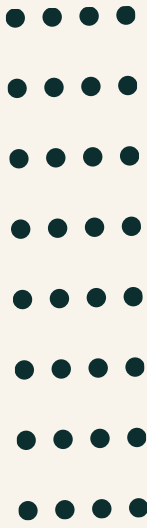
	Dim	Best	Mean	Worst	Success rate (%)
Rastrigin	10	0	0.000001	0.000002	100
	30	0.000054	0.000097	0.000158	100
	50	0.000317	0.00054	0.000762	100
 Styblinski-Tang -39.16617 * N	10	-352.328	-324.111	-299.306	0
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	50	-908.881	-847.778	-791.056	0
 Griewank	10	0	0.006611	0.041062	90
	30	0	0	0	100
	50	0	0	0.000001	100





RESULT-HYBRID(EPOCH0.3 ACTIVE)

	Dim	Best	Mean	Worst	Success rate (%)
Rastrigin	10	0	0.000001	0.000002	100
	30	0.000051	0.000075	0.000095	100
	50	0.000245	0.000523	0.000795	100
 Styblinski-Tang -39.16617 * N	10	-363.388	-336.013	-302.453	0
	30	-760.387	-623.297	-516.437	0
	50	-1146.25	-970.54	-843.805	0
Griewank	10	0	0.006559	0.041142	90
	30	0	0	0	100
	50	0	0.000001	0.000001	100





THANK
YOU

